3. First bud production

Contents

1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for survival analysis.

```
FB_df <-

  #' Load the raw data frame.
  df %>%

  #' Define the follow-up time for each individual:
  #' if the hydra produced a first bud -> time of first bud production (event),
  #' if the hydra died -> time of death (censor),
  #' if the hydra did not produce a first bud and stayed alive
  #' -> 42 days (no event, censor at the end of the experiment).
  mutate(age_followup_days = case_when(
    FB_state == 1 ~ age_FB,
    FB_state == 0 & death_state == 1 ~ age_death,
    TRUE ~ 42),
```

```
    age_followup = age_followup_days / 7,
    FB_event = FB_state) %>%

  #' Build a competing-risk variable for the cumulative incidence functions:
  #' if the hydra did not produce a first bud and stayed alive -> 0 (no event),
  #' if the hydra produced a first bud -> 1 (event),
  #' if the hydra died -> 2 (competing risk of death).
  mutate(status_competing = factor(case_when(FB_event == 1 ~ 1,
                                             death_state == 1 ~ 2,
                                             TRUE ~ 0))) %>%

  #' Build the variable tum_state_early: an individual is considered
  #' tumoral only if the tumor appeared before or at the same time as the
  #' supernumerary tentacle event.
  mutate(age_tum_weeks = age_tum / 7,
         tum_state_early = case_when(tum_state == "Tumoral" &
                                       !is.na(age_tum_weeks) &
                                       age_tum_weeks <= age_followup ~ "Tumoral",
                                     TRUE ~ "Healthy"),
         tum_state_early = factor(tum_state_early, levels = c("Healthy", "Tumoral"))) %>%

  #' Remove missing data.
  filter(!is.na(age_followup)) %>%

  #' Keep only the variables needed for the analysis.
  select(age_followup, FB_event, status_competing, sleep_condition, lineage,
  tum_state_early, parent_unique, replicate_unique)
```

Set reference levels for the categorical variables.

```
FB_df$lineage <- relevel(factor(FB_df$lineage), ref = "HO_MT")
FB_df$sleep_condition <- relevel(factor(FB_df$sleep_condition), ref = "Control")
```

3 Survival object

Create a survival object with follow-up time and event indicator.

```
FB_surv_object <- Surv(FB_df$age_followup, FB_df$FB_event)
```

4 Proportional hazards assumption check : Schoenfeld residuals

Fit a preliminary full Cox model (fixed effects only).

```
FB_cox <- coxph(Surv(age_followup, FB_event) ~ sleep_condition * lineage *
tum_state_early, data = FB_df)
```

Test the proportional hazards assumption.

```r
print(cox.zph(FB_cox))
```

```
##                                          chisq df       p
## sleep_condition                           5.55  1   0.019
## lineage                                 131.26  5 < 2e-16
## tum_state_early                           4.39  1   0.036
## sleep_condition:lineage                  55.59  5 9.9e-11
## sleep_condition:tum_state_early           6.19  1   0.013
## lineage:tum_state_early                   5.44  2   0.066
## sleep_condition:lineage:tum_state_early   2.50  1   0.114
## GLOBAL                                  176.70 16 < 2e-16
```

The proportional hazards assumption is violated for at least one term, an accelerated failure time mixed-effects model will be used.

5 AIC comparison of different distributions

Compare intercept-only accelerated failure time models fitted with different parametric distributions to identify the best-fitting one.

```r
FB_dist_weib <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "weibull")

FB_dist_lnorm <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                           dist = "lognormal")

FB_dist_llog <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "loglogistic")

FB_dist_exp <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                         dist = "exponential")

AIC(FB_dist_weib, FB_dist_lnorm, FB_dist_llog, FB_dist_exp) %>% arrange(AIC)
```

```
##               df      AIC
## FB_dist_lnorm  2 2521.574
## FB_dist_llog   2 2530.357
## FB_dist_weib   2 2634.431
## FB_dist_exp    1 2648.340
```

The model with a log-normal distribution has the lowest AIC, we'll use it for the models.

6 Model selection

6.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```r
FB_R_0 <- SurvregME(Surv(age_followup, FB_event) ~
                      sleep_condition * lineage * tum_state_early,
                    data = FB_df, dist = "lognormal")

FB_R_1 <- SurvregME(Surv(age_followup, FB_event) ~
                      sleep_condition * lineage * tum_state_early +
                      (1 | parent_unique),
                    data = FB_df, dist = "lognormal")

FB_R_2 <- SurvregME(Surv(age_followup, FB_event) ~
                      sleep_condition * lineage * tum_state_early +
                      (1 | replicate_unique),
                    data = FB_df, dist = "lognormal")

FB_R_3 <- SurvregME(Surv(age_followup, FB_event) ~
                      sleep_condition * lineage * tum_state_early +
                      (1 | parent_unique) + (1 | replicate_unique),
                    data = FB_df, dist = "lognormal")

AIC(FB_R_0, FB_R_1, FB_R_2, FB_R_3) %>% arrange(AIC)
```

```
##        df      AIC
## FB_R_2 26 2041.720
## FB_R_3 27 2043.720
## FB_R_0 25 2197.253
## FB_R_1 26 2199.253
```

The model with the replicate id as a random effect (FB_R_2) has the lowest AIC, this random effect structure is therefore selected.

6.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```r
FB_F_0 <- SurvregME(Surv(age_followup, FB_event) ~ 1 + (1 | replicate_unique), data =
FB_df, dist = "lognormal")
FB_F_1 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_2 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + (1 | replicate_unique), data
= FB_df, dist = "lognormal")
FB_F_3 <- SurvregME(Surv(age_followup, FB_event) ~ tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_4 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition + lineage + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_5 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition + tum_state_early + (1
| replicate_unique), data = FB_df, dist = "lognormal")
FB_F_6 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_7 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition * lineage + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_8 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition * tum_state_early + (1
| replicate_unique), data = FB_df, dist = "lognormal")
```

```r
FB_F_9 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_10 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition + lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_11 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition * lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_12 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition * tum_state_early +
lineage + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_13 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early +
sleep_condition + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_14 <- SurvregME(Surv(age_followup, FB_event) ~ sleep_condition * lineage *
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")

AIC(FB_F_0, FB_F_1, FB_F_2, FB_F_3, FB_F_4, FB_F_5, FB_F_6, FB_F_7,
    FB_F_8, FB_F_9, FB_F_10, FB_F_11, FB_F_12, FB_F_13, FB_F_14) %>% arrange(AIC)
```

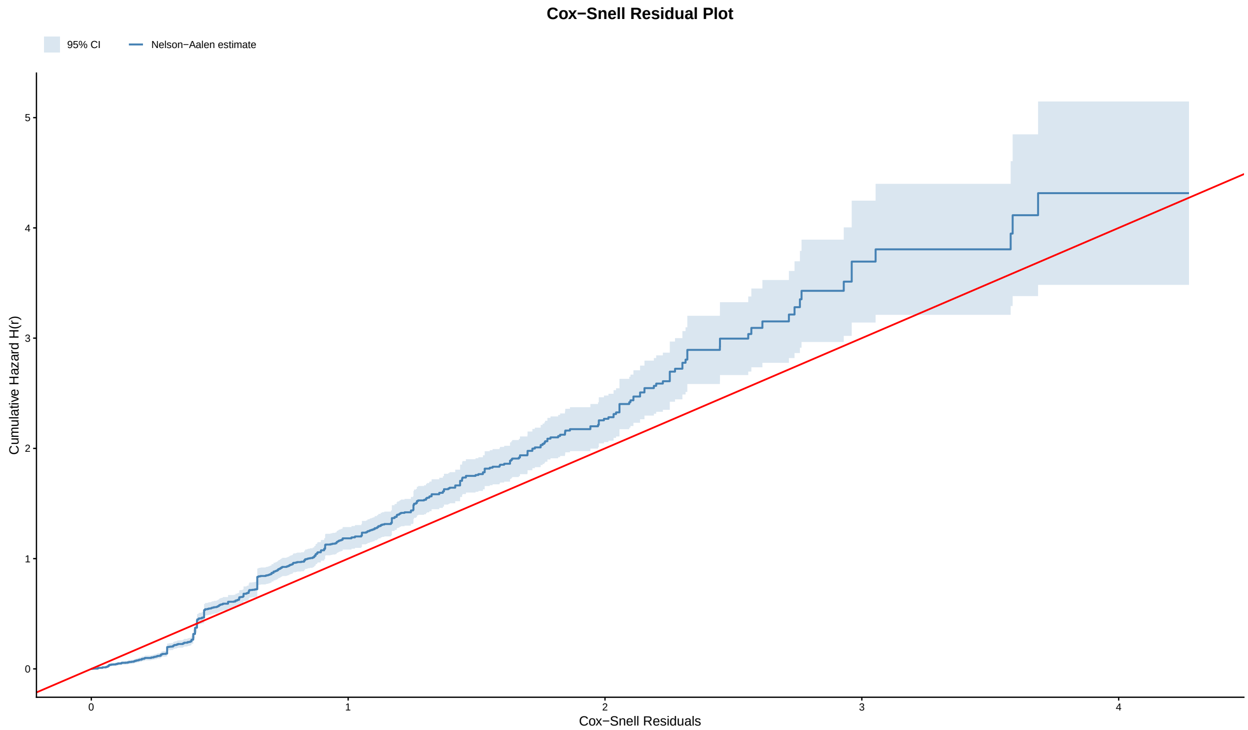
```
##         df      AIC
## FB_F_11 15 2022.674
## FB_F_12 11 2039.850
## FB_F_10 10 2040.381
## FB_F_14 26 2041.720
## FB_F_6   9 2044.891
## FB_F_13 15 2047.613
## FB_F_9  14 2051.542
## FB_F_7  14 2114.692
## FB_F_4   9 2123.404
## FB_F_2   8 2132.611
## FB_F_8   6 2207.268
## FB_F_5   5 2209.009
## FB_F_3   4 2214.100
## FB_F_1   4 2266.732
## FB_F_0   3 2275.753
```

The model with the interaction between the sleep condition and the lineage, and with the additive effect of tumoral state has the lowest AIC and is therefore selected (FB_F_11).

6.3 Goodness-of-fit evaluation

Assess goodness-of-fit of the selected model using Cox-Snell residuals.

```r
plot_cox_snell_survregme(model = FB_F_11, data = FB_df, time_col  = "age_followup",
event_col = "FB_event")
```



The curve follows a 45 degree exponential, we validate the fit of the selected model.

6.4 Selected model summary

#' Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),

                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
##                                                  Estimate Std. Error z value
## sleep_conditionSleep deprivation                 0.048981   0.170051  0.2880
## lineageHC_MT                                     2.211147   0.298500  7.4075
## lineageHO_SPC                                    0.185987   0.257831  0.7214
## lineageHO_SPT                                    1.474533   0.266992  5.5228
```

```
## lineageHO_VLN                                   2.069535   0.252228  8.2050
## lineageHV_GAL                                   1.971254   0.267067  7.3811
## tum_state_earlyTumoral                          2.234789   0.234979  9.5106
## sleep_conditionSleep deprivation:lineageHC_MT   0.578781   0.269384  2.1485
## sleep_conditionSleep deprivation:lineageHO_SPC -0.036154   0.238204 -0.1518
## sleep_conditionSleep deprivation:lineageHO_SPT -0.516544   0.242203 -2.1327
## sleep_conditionSleep deprivation:lineageHO_VLN  0.547600   0.225972  2.4233
## sleep_conditionSleep deprivation:lineageHV_GAL  0.212865   0.239548  0.8886
##                                                 Pr(>|z|)
## sleep_conditionSleep deprivation                 0.77332
## lineageHC_MT                                   1.287e-13 ***
## lineageHO_SPC                                    0.47069
## lineageHO_SPT                                  3.337e-08 ***
## lineageHO_VLN                                  2.306e-16 ***
## lineageHV_GAL                                  1.570e-13 ***
## tum_state_earlyTumoral                         < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHC_MT    0.03167 *
## sleep_conditionSleep deprivation:lineageHO_SPC   0.87936
## sleep_conditionSleep deprivation:lineageHO_SPT   0.03295 *
## sleep_conditionSleep deprivation:lineageHO_VLN   0.01538 *
## sleep_conditionSleep deprivation:lineageHV_GAL   0.37421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

```r
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPC")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),

                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
```

```
##                                                  Estimate Std. Error z value
## sleep_conditionSleep deprivation                 0.012831   0.167869  0.0764
## lineageHO_MT                                    -0.185979   0.257831 -0.7213
## lineageHC_MT                                     2.025164   0.296385  6.8329
## lineageHO_SPT                                    1.288545   0.265838  4.8471
## lineageHO_VLN                                    1.883548   0.249845  7.5389
## lineageHV_GAL                                    1.785271   0.264853  6.7406
## tum_state_earlyTumoral                           2.234794   0.234979  9.5106
## sleep_conditionSleep deprivation:lineageHO_MT    0.036143   0.238204  0.1517
## sleep_conditionSleep deprivation:lineageHC_MT    0.614928   0.267964  2.2948
## sleep_conditionSleep deprivation:lineageHO_SPT -0.480394   0.240519 -1.9973
## sleep_conditionSleep deprivation:lineageHO_VLN  0.583752   0.224262  2.6030
## sleep_conditionSleep deprivation:lineageHV_GAL  0.249012   0.237980  1.0464
##                                                  Pr(>|z|)
## sleep_conditionSleep deprivation                 0.939071
## lineageHO_MT                                     0.470712
## lineageHC_MT                                    8.322e-12 ***
## lineageHO_SPT                                   1.253e-06 ***
## lineageHO_VLN                                   4.741e-14 ***
## lineageHV_GAL                                   1.577e-11 ***
## tum_state_earlyTumoral                          < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHO_MT    0.879400
## sleep_conditionSleep deprivation:lineageHC_MT    0.021744 *
## sleep_conditionSleep deprivation:lineageHO_SPT  0.045790 *
## sleep_conditionSleep deprivation:lineageHO_VLN  0.009242 **
## sleep_conditionSleep deprivation:lineageHV_GAL  0.295396
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

```r
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPT")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),

                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
```

```
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
##                                              Estimate Std. Error z value
## sleep_conditionSleep deprivation            -0.46757    0.17350 -2.6949
## lineageHO_SPC                               -1.28855    0.26584 -4.8471
## lineageHO_MT                                -1.47453    0.26699 -5.5228
## lineageHC_MT                                 0.73661    0.29238  2.5194
## lineageHO_VLN                                0.59499    0.24624  2.4163
## lineageHV_GAL                                0.49671    0.26189  1.8966
## tum_state_earlyTumoral                       2.23480    0.23498  9.5106
## sleep_conditionSleep deprivation:lineageHO_SPC  0.48040    0.24052  1.9974
## sleep_conditionSleep deprivation:lineageHO_MT   0.51654    0.24220  2.1327
## sleep_conditionSleep deprivation:lineageHC_MT   1.09533    0.27264  4.0176
## sleep_conditionSleep deprivation:lineageHO_VLN  1.06416    0.22974  4.6319
## sleep_conditionSleep deprivation:lineageHV_GAL  0.72942    0.24241  3.0090
##                                               Pr(>|z|)
## sleep_conditionSleep deprivation              0.007041 **
## lineageHO_SPC                                1.253e-06 ***
## lineageHO_MT                                 3.337e-08 ***
## lineageHC_MT                                  0.011756 *
## lineageHO_VLN                                 0.015678 *
## lineageHV_GAL                                 0.057877 .
## tum_state_earlyTumoral                       < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHO_SPC  0.045787 *
## sleep_conditionSleep deprivation:lineageHO_MT   0.032950 *
## sleep_conditionSleep deprivation:lineageHC_MT  5.880e-05 ***
## sleep_conditionSleep deprivation:lineageHO_VLN 3.623e-06 ***
## sleep_conditionSleep deprivation:lineageHV_GAL  0.002621 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

```r
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_VLN")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),

                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
```

```
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
##                                                 Estimate Std. Error z value
## sleep_conditionSleep deprivation                0.596595   0.149017  4.0035
## lineageHO_SPT                                  -0.594993   0.246240 -2.4163
## lineageHO_SPC                                  -1.883540   0.249845 -7.5388
## lineageHO_MT                                   -2.069524   0.252228 -8.2050
## lineageHC_MT                                    0.141629   0.272716  0.5193
## lineageHV_GAL                                  -0.098264   0.240411 -0.4087
## tum_state_earlyTumoral                          2.234804   0.234979  9.5107
## sleep_conditionSleep deprivation:lineageHO_SPT -1.064162   0.229745 -4.6319
## sleep_conditionSleep deprivation:lineageHO_SPC -0.583767   0.224262 -2.6031
## sleep_conditionSleep deprivation:lineageHO_MT  -0.547622   0.225972 -2.4234
## sleep_conditionSleep deprivation:lineageHC_MT   0.031167   0.255372  0.1220
## sleep_conditionSleep deprivation:lineageHV_GAL -0.334754   0.224607 -1.4904
##                                                 Pr(>|z|)
## sleep_conditionSleep deprivation               6.240e-05 ***
## lineageHO_SPT                                    0.01568 *
## lineageHO_SPC                                  4.742e-14 ***
## lineageHO_MT                                   2.307e-16 ***
## lineageHC_MT                                     0.60353
## lineageHV_GAL                                    0.68273
## tum_state_earlyTumoral                         < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHO_SPT 3.623e-06 ***
## sleep_conditionSleep deprivation:lineageHO_SPC   0.00924 **
## sleep_conditionSleep deprivation:lineageHO_MT    0.01538 *
## sleep_conditionSleep deprivation:lineageHC_MT    0.90286
## sleep_conditionSleep deprivation:lineageHV_GAL   0.13612
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HC_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),
```

```r
                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
##                                                 Estimate Std. Error z value
## sleep_conditionSleep deprivation                0.627755   0.209080  3.0025
## lineageHO_VLN                                  -0.141603   0.272716 -0.5192
## lineageHO_SPT                                  -0.736598   0.292376 -2.5194
## lineageHO_SPC                                  -2.025149   0.296384 -6.8329
## lineageHO_MT                                   -2.211131   0.298500 -7.4075
## lineageHV_GAL                                  -0.239892   0.287121 -0.8355
## tum_state_earlyTumoral                          2.234790   0.234979  9.5106
## sleep_conditionSleep deprivation:lineageHO_VLN -0.031172   0.255372 -0.1221
## sleep_conditionSleep deprivation:lineageHO_SPT -1.095320   0.272637 -4.0175
## sleep_conditionSleep deprivation:lineageHO_SPC -0.614927   0.267964 -2.2948
## sleep_conditionSleep deprivation:lineageHO_MT  -0.578778   0.269384 -2.1485
## sleep_conditionSleep deprivation:lineageHV_GAL -0.365902   0.268223 -1.3642
##                                                 Pr(>|z|)
## sleep_conditionSleep deprivation                0.002678 **
## lineageHO_VLN                                   0.603600
## lineageHO_SPT                                   0.011757 *
## lineageHO_SPC                                  8.324e-12 ***
## lineageHO_MT                                   1.287e-13 ***
## lineageHV_GAL                                   0.403433
## tum_state_earlyTumoral                         < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHO_VLN  0.902847
## sleep_conditionSleep deprivation:lineageHO_SPT 5.882e-05 ***
## sleep_conditionSleep deprivation:lineageHO_SPC  0.021744 *
## sleep_conditionSleep deprivation:lineageHO_MT   0.031672 *
## sleep_conditionSleep deprivation:lineageHV_GAL  0.172514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

```r
FB_df$lineage <- relevel(FB_df$lineage, ref = "HV_GAL")
summary(SurvregME(Surv(age_followup, FB_event) ~
                    sleep_condition * lineage + tum_state_early + (1 | replicate_unique),

                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
##  Formula:
## Surv(age_followup, FB_event) ~ sleep_condition * lineage + tum_state_early +
##     (1 | replicate_unique)
##
##  Fitted to dataset FB_df
##
##  Fixed effects parameters:
##  =========================
##
##                                                 Estimate Std. Error z value
## sleep_conditionSleep deprivation                0.261860   0.168784  1.5515
## lineageHC_MT                                    0.239910   0.287122  0.8356
## lineageHO_VLN                                   0.098294   0.240411  0.4089
## lineageHO_SPT                                  -0.496709   0.261894 -1.8966
## lineageHO_SPC                                  -1.785257   0.264853 -6.7406
## lineageHO_MT                                   -1.971238   0.267067 -7.3811
## tum_state_earlyTumoral                          2.234796   0.234979  9.5106
## sleep_conditionSleep deprivation:lineageHC_MT   0.365898   0.268223  1.3642
## sleep_conditionSleep deprivation:lineageHO_VLN  0.334720   0.224607  1.4902
## sleep_conditionSleep deprivation:lineageHO_SPT -0.729426   0.242415 -3.0090
## sleep_conditionSleep deprivation:lineageHO_SPC -0.249033   0.237980 -1.0464
## sleep_conditionSleep deprivation:lineageHO_MT  -0.212888   0.239548 -0.8887
##                                                 Pr(>|z|)
## sleep_conditionSleep deprivation                0.120793
## lineageHC_MT                                    0.403398
## lineageHO_VLN                                   0.682642
## lineageHO_SPT                                   0.057880 .
## lineageHO_SPC                                  1.578e-11 ***
## lineageHO_MT                                   1.570e-13 ***
## tum_state_earlyTumoral                         < 2.2e-16 ***
## sleep_conditionSleep deprivation:lineageHC_MT   0.172519
## sleep_conditionSleep deprivation:lineageHO_VLN  0.136159
## sleep_conditionSleep deprivation:lineageHO_SPT  0.002621 **
## sleep_conditionSleep deprivation:lineageHO_SPC  0.295356
## sleep_conditionSleep deprivation:lineageHO_MT   0.374160
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  Random effects:
##  ===============
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
```

```
##      1.1703
##
##
##  Log-likelihood: -996.3371 (npar = 15)
```

7 Median times

7.1 By sleep condition

Estimate cumulative incidence functions for the first bud event, accounting for the competing risk of death, stratified by sleep condition.

```
FB_cuminc_sleep <- cuminc(Surv(age_followup, status_competing) ~ sleep_condition, data =
FB_df)
```

For each stratum, extract the median time to event and its 95CI.

```
FB_medians_sleep <- tidy(FB_cuminc_sleep) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_sleep)
```

```
## # A tibble: 2 x 4
##   strata            median_time ci_low ci_high
##   <fct>                   <dbl>  <dbl>   <dbl>
## 1 Control                  1.14   1.14    1.14
## 2 Sleep deprivation        1.14   1.14    1.29
```

7.2 By lineage

Same method as above, but stratified by lineage.

```
FB_cuminc_lineage <- cuminc(Surv(age_followup, status_competing) ~ lineage, data = FB_df)

FB_medians_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
```

```
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_lineage)
```

```
## # A tibble: 6 x 4
##   strata median_time ci_low ci_high
##   <fct>        <dbl>  <dbl>   <dbl>
## 1 HV_GAL       1.71   1.29     2
## 2 HC_MT        3      2.43     3
## 3 HO_VLN       1      1        2.43
## 4 HO_SPT       1.29   1.14     1.29
## 5 HO_SPC       0.286  0.286    1.14
## 6 HO_MT        0.286  0.286    1
```

7.3 By sleep condition and lineage

Same method as above, but stratified by sleep condition and lineage.

```
FB_cuminc_interaction <- cuminc(Surv(age_followup, status_competing) ~ sleep_condition +
lineage, data = FB_df)

FB_medians_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_interaction)
```

```
## # A tibble: 12 x 4
##    strata                     median_time ci_low ci_high
##    <fct>                            <dbl>  <dbl>   <dbl>
##  1 Control, HV_GAL                  1.71   1.71     2
##  2 Sleep deprivation, HV_GAL        1.71   1        2.43
##  3 Control, HC_MT                   1.43   1.14     3
##  4 Sleep deprivation, HC_MT         3      3        3
##  5 Control, HO_VLN                  1      1        1.14
##  6 Sleep deprivation, HO_VLN        1.14   1        3.14
##  7 Control, HO_SPT                  1.29   1.14     1.57
##  8 Sleep deprivation, HO_SPT        1.14   1.14     1.29
```

```
##  9 Control, HO_SPC                  0.286  0.286    1.14
## 10 Sleep deprivation, HO_SPC        0.286  0.286    1.14
## 11 Control, HO_MT                   0.286  0.286    1
## 12 Sleep deprivation, HO_MT         0.286  0.286    1
```

7.4 By tumoral state

Same method as above, but stratified by tumoral state.

```
FB_cuminc_tum <- cuminc(Surv(age_followup, status_competing) ~ tum_state_early, data =
FB_df)

FB_medians_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_tum)
```

```
## # A tibble: 2 x 4
##   strata  median_time ci_low ci_high
##   <fct>         <dbl>  <dbl>   <dbl>
## 1 Healthy        1.14   1.14    1.14
## 2 Tumoral        3      2.29   NA
```

8 Cumulative incidences

8.1 By week and sleep condition

Convert the continuous cumulative incidence function into a weekly summary table: for each stratum of sleep condition and each week, show the estimate with its 95CI.

```
FB_incidence_sleep <- tidy(FB_cuminc_sleep) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_sleep)
```

```
## # A tibble: 14 x 5
##    strata             week estimate_cuminc conf.low conf.high
##    <fct>             <dbl>           <dbl>    <dbl>     <dbl>
##  1 Control               0           0       NA        NA
##  2 Control               1           0.433    0.386     0.479
##  3 Control               2           0.736    0.692     0.775
##  4 Control               3           0.894    0.860     0.919
##  5 Control               4           0.907    0.876     0.931
##  6 Control               5           0.931    0.902     0.951
##  7 Control               6           0.947    0.921     0.964
##  8 Sleep deprivation     0           0       NA        NA
##  9 Sleep deprivation     1           0.421    0.374     0.467
## 10 Sleep deprivation     2           0.650    0.603     0.693
## 11 Sleep deprivation     3           0.789    0.748     0.825
## 12 Sleep deprivation     4           0.822    0.782     0.855
## 13 Sleep deprivation     5           0.880    0.845     0.907
## 14 Sleep deprivation     6           0.882    0.848     0.909
```

8.2 By week and lineage

Same method as above, but stratified by lineage.

```
FB_incidence_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_lineage)
```

```
## # A tibble: 42 x 5
##    strata  week estimate_cuminc conf.low conf.high
##    <fct>  <dbl>           <dbl>    <dbl>     <dbl>
##  1 HV_GAL     0          0       NA        NA
##  2 HV_GAL     1          0.319    0.245     0.396
##  3 HV_GAL     2          0.597    0.512     0.672
##  4 HV_GAL     3          0.868    0.800     0.914
##  5 HV_GAL     4          0.875    0.808     0.920
##  6 HV_GAL     5          0.931    0.874     0.962
##  7 HV_GAL     6          0.938    0.882     0.967
##  8 HC_MT      0          0       NA        NA
##  9 HC_MT      1          0.0729   0.0320    0.137
## 10 HC_MT      2          0.375    0.279     0.471
## 11 HC_MT      3          0.844    0.754     0.903
## 12 HC_MT      4          0.917    0.838     0.958
## 13 HC_MT      5          0.938    0.862     0.972
## 14 HC_MT      6          0.938    0.862     0.972
## 15 HO_VLN     0          0       NA        NA
## 16 HO_VLN     1          0.536    0.463     0.604
```

```
## 17 HO_VLN     2          0.573    0.500      0.640
## 18 HO_VLN     3          0.672    0.600      0.734
## 19 HO_VLN     4          0.703    0.633      0.763
## 20 HO_VLN     5          0.813    0.749      0.861
## 21 HO_VLN     6          0.849    0.789      0.893
## 22 HO_SPT     0          0       NA         NA
## 23 HO_SPT     1          0.215    0.152      0.286
## 24 HO_SPT     2          0.819    0.746      0.874
## 25 HO_SPT     3          0.868    0.800      0.914
## 26 HO_SPT     4          0.882    0.816      0.925
## 27 HO_SPT     5          0.882    0.816      0.925
## 28 HO_SPT     6          0.882    0.816      0.925
## 29 HO_SPC     0          0       NA         NA
## 30 HO_SPC     1          0.549    0.463      0.626
## 31 HO_SPC     2          0.882    0.816      0.925
## 32 HO_SPC     3          0.924    0.865      0.957
## 33 HO_SPC     4          0.951    0.899      0.977
## 34 HO_SPC     5          0.979    0.931      0.994
## 35 HO_SPC     6          0.979    0.931      0.994
## 36 HO_MT      0          0       NA         NA
## 37 HO_MT      1          0.715    0.634      0.781
## 38 HO_MT      2          0.847    0.778      0.896
## 39 HO_MT      3          0.931    0.874      0.962
## 40 HO_MT      4          0.931    0.874      0.962
## 41 HO_MT      5          0.931    0.874      0.962
## 42 HO_MT      6          0.931    0.874      0.962
```

8.3 By week, sleep condition, and lineage

Same method as above, but stratified by sleep condition and lineage.

```
FB_incidence_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_interaction)
```

```
## # A tibble: 84 x 5
##    strata                     week estimate_cuminc conf.low conf.high
##    <fct>                     <dbl>           <dbl>    <dbl>     <dbl>
##  1 Control, HV_GAL               0           0       NA        NA
##  2 Control, HV_GAL               1           0.222    0.134     0.324
##  3 Control, HV_GAL               2           0.639    0.515     0.739
##  4 Control, HV_GAL               3           0.986    0.898     0.998
##  5 Control, HV_GAL               4          NA       NA        NA
##  6 Control, HV_GAL               5          NA       NA        NA
##  7 Control, HV_GAL               6          NA       NA        NA
```

```
## 8 Sleep deprivation, HV_GAL     0           0       NA          NA
## 9 Sleep deprivation, HV_GAL     1           0.417    0.302       0.528
## 10 Sleep deprivation, HV_GAL    2           0.556    0.432       0.662
## 11 Sleep deprivation, HV_GAL    3           0.75     0.631       0.835
## 12 Sleep deprivation, HV_GAL    4           0.75     0.631       0.835
## 13 Sleep deprivation, HV_GAL    5           0.861    0.755       0.924
## 14 Sleep deprivation, HV_GAL    6           0.875    0.771       0.934
## 15 Control, HC_MT               0           0       NA          NA
## 16 Control, HC_MT               1           0.0833   0.0263      0.183
## 17 Control, HC_MT               2           0.521    0.370       0.652
## 18 Control, HC_MT               3           0.917    0.788       0.969
## 19 Control, HC_MT               4           0.979    0.777       0.998
## 20 Control, HC_MT               5           0.979    0.777       0.998
## 21 Control, HC_MT               6           0.979    0.777       0.998
## 22 Sleep deprivation, HC_MT     0           0       NA          NA
## 23 Sleep deprivation, HC_MT     1           0.0625   0.0160      0.156
## 24 Sleep deprivation, HC_MT     2           0.229    0.122       0.356
## 25 Sleep deprivation, HC_MT     3           0.771    0.623       0.867
## 26 Sleep deprivation, HC_MT     4           0.854    0.713       0.929
## 27 Sleep deprivation, HC_MT     5           0.896    0.759       0.957
## 28 Sleep deprivation, HC_MT     6           0.896    0.759       0.957
## 29 Control, HO_VLN              0           0       NA          NA
## 30 Control, HO_VLN              1           0.604    0.499       0.694
## 31 Control, HO_VLN              2           0.635    0.530       0.723
## 32 Control, HO_VLN              3           0.75     0.650       0.825
## 33 Control, HO_VLN              4           0.771    0.672       0.843
## 34 Control, HO_VLN              5           0.865    0.776       0.920
## 35 Control, HO_VLN              6           0.938    0.863       0.972
## 36 Sleep deprivation, HO_VLN    0           0       NA          NA
## 37 Sleep deprivation, HO_VLN    1           0.469    0.366       0.565
## 38 Sleep deprivation, HO_VLN    2           0.510    0.406       0.606
## 39 Sleep deprivation, HO_VLN    3           0.594    0.488       0.685
## 40 Sleep deprivation, HO_VLN    4           0.635    0.530       0.723
## 41 Sleep deprivation, HO_VLN    5           0.760    0.661       0.835
## 42 Sleep deprivation, HO_VLN    6           0.760    0.661       0.835
## 43 Control, HO_SPT              0           0       NA          NA
## 44 Control, HO_SPT              1           0.194    0.112       0.293
## 45 Control, HO_SPT              2           0.819    0.707       0.892
## 46 Control, HO_SPT              3           0.847    0.738       0.914
## 47 Control, HO_SPT              4           0.847    0.738       0.914
## 48 Control, HO_SPT              5           0.847    0.738       0.914
## 49 Control, HO_SPT              6           0.847    0.738       0.914
## 50 Sleep deprivation, HO_SPT    0           0       NA          NA
## 51 Sleep deprivation, HO_SPT    1           0.236    0.145       0.340
## 52 Sleep deprivation, HO_SPT    2           0.819    0.707       0.892
## 53 Sleep deprivation, HO_SPT    3           0.889    0.786       0.944
## 54 Sleep deprivation, HO_SPT    4           0.917    0.818       0.963
## 55 Sleep deprivation, HO_SPT    5           0.917    0.818       0.963
## 56 Sleep deprivation, HO_SPT    6           0.917    0.818       0.963
## 57 Control, HO_SPC              0           0       NA          NA
## 58 Control, HO_SPC              1           0.542    0.419       0.649
## 59 Control, HO_SPC              2           0.889    0.787       0.944
## 60 Control, HO_SPC              3           0.972    0.870       0.994
## 61 Control, HO_SPC              4           0.972    0.870       0.994
```

```
## 62 Control, HO_SPC              5          NA       NA          NA
## 63 Control, HO_SPC              6          NA       NA          NA
## 64 Sleep deprivation, HO_SPC    0           0       NA          NA
## 65 Sleep deprivation, HO_SPC    1           0.556    0.433       0.662
## 66 Sleep deprivation, HO_SPC    2           0.875    0.771       0.934
## 67 Sleep deprivation, HO_SPC    3           0.875    0.771       0.934
## 68 Sleep deprivation, HO_SPC    4           0.931    0.836       0.972
## 69 Sleep deprivation, HO_SPC    5           0.972    0.874       0.994
## 70 Sleep deprivation, HO_SPC    6           0.972    0.874       0.994
## 71 Control, HO_MT               0           0       NA          NA
## 72 Control, HO_MT               1           0.778    0.663       0.858
## 73 Control, HO_MT               2           0.875    0.771       0.934
## 74 Control, HO_MT               3           0.944    0.852       0.980
## 75 Control, HO_MT               4           0.944    0.852       0.980
## 76 Control, HO_MT               5           0.944    0.852       0.980
## 77 Control, HO_MT               6           0.944    0.852       0.980
## 78 Sleep deprivation, HO_MT     0           0       NA          NA
## 79 Sleep deprivation, HO_MT     1           0.653    0.530       0.751
## 80 Sleep deprivation, HO_MT     2           0.819    0.709       0.891
## 81 Sleep deprivation, HO_MT     3           0.917    0.821       0.962
## 82 Sleep deprivation, HO_MT     4           0.917    0.821       0.962
## 83 Sleep deprivation, HO_MT     5           0.917    0.821       0.962
## 84 Sleep deprivation, HO_MT     6           0.917    0.821       0.962
```

8.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```r
FB_incidence_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_tum)
```

```
## # A tibble: 14 x 5
##    strata   week estimate_cuminc conf.low conf.high
##    <fct>   <dbl>           <dbl>    <dbl>     <dbl>
##  1 Healthy     0          0       NA          NA
##  2 Healthy     1          0.447    0.413       0.481
##  3 Healthy     2          0.712    0.680       0.742
##  4 Healthy     3          0.858    0.832       0.880
##  5 Healthy     4          0.880    0.856       0.900
##  6 Healthy     5          0.919    0.898       0.935
##  7 Healthy     6          0.928    0.908       0.944
##  8 Tumoral     0          0       NA          NA
##  9 Tumoral     1          0.0244   0.00185     0.112
## 10 Tumoral     2          0.317    0.181       0.462
```

```
## 11 Tumoral     3          0.512   0.349       0.654
## 12 Tumoral     4          0.561   0.394       0.699
## 13 Tumoral     5          0.634   0.464       0.763
## 14 Tumoral     6          0.634   0.464       0.763
```

9 Cumulative proportions

9.1 By week and sleep condition

Compute raw cumulative proportions of first bud production per week, stratified by sleep condition.

```r
FB_proportions_sleep <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(sleep_condition) %>%
  dplyr::summarise(total_N = n(),
                   n1 = sum(week_FB_event <= 1 & FB_event == 1),
                   n2 = sum(week_FB_event <= 2 & FB_event == 1),
                   n3 = sum(week_FB_event <= 3 & FB_event == 1),
                   n4 = sum(week_FB_event <= 4 & FB_event == 1),
                   n5 = sum(week_FB_event <= 5 & FB_event == 1),
                   n6 = sum(week_FB_event <= 6 & FB_event == 1),
                   p1 = n1 / total_N,
                   p2 = n2 / total_N,
                   p3 = n3 / total_N,
                   p4 = n4 / total_N,
                   p5 = n5 / total_N,
                   p6 = n6 / total_N,
                   .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^[np][1-6]$"),
                      names_to = c(".value", "week"),
                      names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(sleep_condition, week, n, total_N, raw_proportion = p)

print(FB_proportions_sleep)
```

```
## # A tibble: 12 x 5
##    sleep_condition    week     n total_N raw_proportion
##    <fct>             <dbl> <int>   <int>          <dbl>
##  1 Control               1   187     432          0.433
##  2 Control               2   318     432          0.736
##  3 Control               3   386     432          0.894
##  4 Control               4   392     432          0.907
##  5 Control               5   402     432          0.931
##  6 Control               6   409     432          0.947
##  7 Sleep deprivation     1   182     432          0.421
##  8 Sleep deprivation     2   281     432          0.650
##  9 Sleep deprivation     3   341     432          0.789
## 10 Sleep deprivation     4   355     432          0.822
## 11 Sleep deprivation     5   380     432          0.880
## 12 Sleep deprivation     6   381     432          0.882
```

9.2 By week and lineage

Same method as above, but stratified by lineage.

```
FB_proportions_lineage <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(lineage) %>%
  dplyr::summarise(total_N = n(),
                   n1 = sum(week_FB_event <= 1 & FB_event == 1),
                   n2 = sum(week_FB_event <= 2 & FB_event == 1),
                   n3 = sum(week_FB_event <= 3 & FB_event == 1),
                   n4 = sum(week_FB_event <= 4 & FB_event == 1),
                   n5 = sum(week_FB_event <= 5 & FB_event == 1),
                   n6 = sum(week_FB_event <= 6 & FB_event == 1),
                   p1 = n1 / total_N,
                   p2 = n2 / total_N,
                   p3 = n3 / total_N,
                   p4 = n4 / total_N,
                   p5 = n5 / total_N,
                   p6 = n6 / total_N,
                   .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^[np][1-6]$"),
                      names_to = c(".value", "week"),
                      names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_lineage)
```

```
## # A tibble: 36 x 5
##    lineage  week     n total_N raw_proportion
##    <fct>   <dbl> <int>   <int>          <dbl>
##  1 HV_GAL      1    46     144         0.319
##  2 HV_GAL      2    86     144         0.597
##  3 HV_GAL      3   125     144         0.868
##  4 HV_GAL      4   126     144         0.875
##  5 HV_GAL      5   134     144         0.931
##  6 HV_GAL      6   135     144         0.938
##  7 HC_MT       1     7      96         0.0729
##  8 HC_MT       2    36      96         0.375
##  9 HC_MT       3    81      96         0.844
## 10 HC_MT       4    88      96         0.917
## 11 HC_MT       5    90      96         0.938
## 12 HC_MT       6    90      96         0.938
## 13 HO_VLN      1   103     192         0.536
## 14 HO_VLN      2   110     192         0.573
## 15 HO_VLN      3   129     192         0.672
## 16 HO_VLN      4   135     192         0.703
## 17 HO_VLN      5   156     192         0.812
## 18 HO_VLN      6   163     192         0.849
## 19 HO_SPT      1    31     144         0.215
## 20 HO_SPT      2   118     144         0.819
## 21 HO_SPT      3   125     144         0.868
## 22 HO_SPT      4   127     144         0.882
```

```
## 23 HO_SPT      5    127      144          0.882
## 24 HO_SPT      6    127      144          0.882
## 25 HO_SPC      1     79      144          0.549
## 26 HO_SPC      2    127      144          0.882
## 27 HO_SPC      3    133      144          0.924
## 28 HO_SPC      4    137      144          0.951
## 29 HO_SPC      5    141      144          0.979
## 30 HO_SPC      6    141      144          0.979
## 31 HO_MT       1    103      144          0.715
## 32 HO_MT       2    122      144          0.847
## 33 HO_MT       3    134      144          0.931
## 34 HO_MT       4    134      144          0.931
## 35 HO_MT       5    134      144          0.931
## 36 HO_MT       6    134      144          0.931
```

9.3 By week, sleep condition, and lineage

Same method as above, but stratified by sleep condition and lineage.

```
FB_proportions_interaction <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(sleep_condition, lineage) %>%
  dplyr::summarise(total_N = n(),
                   n1 = sum(week_FB_event <= 1 & FB_event == 1),
                   n2 = sum(week_FB_event <= 2 & FB_event == 1),
                   n3 = sum(week_FB_event <= 3 & FB_event == 1),
                   n4 = sum(week_FB_event <= 4 & FB_event == 1),
                   n5 = sum(week_FB_event <= 5 & FB_event == 1),
                   n6 = sum(week_FB_event <= 6 & FB_event == 1),
                   p1 = n1 / total_N,
                   p2 = n2 / total_N,
                   p3 = n3 / total_N,
                   p4 = n4 / total_N,
                   p5 = n5 / total_N,
                   p6 = n6 / total_N,
                   .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^[np][1-6]$"),
                      names_to = c(".value", "week"),
                      names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(sleep_condition, lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_interaction)
```

```
## # A tibble: 72 x 6
##    sleep_condition   lineage  week     n total_N raw_proportion
##    <fct>             <fct>   <dbl> <int>   <int>          <dbl>
##  1 Control           HV_GAL      1    16      72          0.222
##  2 Control           HV_GAL      2    46      72          0.639
##  3 Control           HV_GAL      3    71      72          0.986
##  4 Control           HV_GAL      4    72      72          1
##  5 Control           HV_GAL      5    72      72          1
##  6 Control           HV_GAL      6    72      72          1
```

```
##  7 Control           HC_MT      1     4      48           0.0833
##  8 Control           HC_MT      2    25      48           0.521
##  9 Control           HC_MT      3    44      48           0.917
## 10 Control           HC_MT      4    47      48           0.979
## 11 Control           HC_MT      5    47      48           0.979
## 12 Control           HC_MT      6    47      48           0.979
## 13 Control           HO_VLN     1    58      96           0.604
## 14 Control           HO_VLN     2    61      96           0.635
## 15 Control           HO_VLN     3    72      96           0.75
## 16 Control           HO_VLN     4    74      96           0.771
## 17 Control           HO_VLN     5    83      96           0.865
## 18 Control           HO_VLN     6    90      96           0.938
## 19 Control           HO_SPT     1    14      72           0.194
## 20 Control           HO_SPT     2    59      72           0.819
## 21 Control           HO_SPT     3    61      72           0.847
## 22 Control           HO_SPT     4    61      72           0.847
## 23 Control           HO_SPT     5    61      72           0.847
## 24 Control           HO_SPT     6    61      72           0.847
## 25 Control           HO_SPC     1    39      72           0.542
## 26 Control           HO_SPC     2    64      72           0.889
## 27 Control           HO_SPC     3    70      72           0.972
## 28 Control           HO_SPC     4    70      72           0.972
## 29 Control           HO_SPC     5    71      72           0.986
## 30 Control           HO_SPC     6    71      72           0.986
## 31 Control           HO_MT      1    56      72           0.778
## 32 Control           HO_MT      2    63      72           0.875
## 33 Control           HO_MT      3    68      72           0.944
## 34 Control           HO_MT      4    68      72           0.944
## 35 Control           HO_MT      5    68      72           0.944
## 36 Control           HO_MT      6    68      72           0.944
## 37 Sleep deprivation HV_GAL     1    30      72           0.417
## 38 Sleep deprivation HV_GAL     2    40      72           0.556
## 39 Sleep deprivation HV_GAL     3    54      72           0.75
## 40 Sleep deprivation HV_GAL     4    54      72           0.75
## 41 Sleep deprivation HV_GAL     5    62      72           0.861
## 42 Sleep deprivation HV_GAL     6    63      72           0.875
## 43 Sleep deprivation HC_MT      1     3      48           0.0625
## 44 Sleep deprivation HC_MT      2    11      48           0.229
## 45 Sleep deprivation HC_MT      3    37      48           0.771
## 46 Sleep deprivation HC_MT      4    41      48           0.854
## 47 Sleep deprivation HC_MT      5    43      48           0.896
## 48 Sleep deprivation HC_MT      6    43      48           0.896
## 49 Sleep deprivation HO_VLN     1    45      96           0.469
## 50 Sleep deprivation HO_VLN     2    49      96           0.510
## 51 Sleep deprivation HO_VLN     3    57      96           0.594
## 52 Sleep deprivation HO_VLN     4    61      96           0.635
## 53 Sleep deprivation HO_VLN     5    73      96           0.760
## 54 Sleep deprivation HO_VLN     6    73      96           0.760
## 55 Sleep deprivation HO_SPT     1    17      72           0.236
## 56 Sleep deprivation HO_SPT     2    59      72           0.819
## 57 Sleep deprivation HO_SPT     3    64      72           0.889
## 58 Sleep deprivation HO_SPT     4    66      72           0.917
## 59 Sleep deprivation HO_SPT     5    66      72           0.917
## 60 Sleep deprivation HO_SPT     6    66      72           0.917
```

```
## 61 Sleep deprivation HO_SPC      1    40      72          0.556
## 62 Sleep deprivation HO_SPC      2    63      72          0.875
## 63 Sleep deprivation HO_SPC      3    63      72          0.875
## 64 Sleep deprivation HO_SPC      4    67      72          0.931
## 65 Sleep deprivation HO_SPC      5    70      72          0.972
## 66 Sleep deprivation HO_SPC      6    70      72          0.972
## 67 Sleep deprivation HO_MT       1    47      72          0.653
## 68 Sleep deprivation HO_MT       2    59      72          0.819
## 69 Sleep deprivation HO_MT       3    66      72          0.917
## 70 Sleep deprivation HO_MT       4    66      72          0.917
## 71 Sleep deprivation HO_MT       5    66      72          0.917
## 72 Sleep deprivation HO_MT       6    66      72          0.917
```

9.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```r
FB_proportions_tum <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(tum_state_early) %>%
  dplyr::summarise(total_N = n(),
                   n1 = sum(week_FB_event <= 1 & FB_event == 1),
                   n2 = sum(week_FB_event <= 2 & FB_event == 1),
                   n3 = sum(week_FB_event <= 3 & FB_event == 1),
                   n4 = sum(week_FB_event <= 4 & FB_event == 1),
                   n5 = sum(week_FB_event <= 5 & FB_event == 1),
                   n6 = sum(week_FB_event <= 6 & FB_event == 1),
                   p1 = n1 / total_N,
                   p2 = n2 / total_N,
                   p3 = n3 / total_N,
                   p4 = n4 / total_N,
                   p5 = n5 / total_N,
                   p6 = n6 / total_N,
                   .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^[np][1-6]$"),
                      names_to = c(".value", "week"),
                      names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(tum_state_early, week, n, total_N, raw_proportion = p)

print(FB_proportions_tum)
```

```
## # A tibble: 12 x 5
##    tum_state_early  week     n total_N raw_proportion
##    <fct>           <dbl> <int>   <int>          <dbl>
##  1 Healthy             1   368     823         0.447
##  2 Healthy             2   586     823         0.712
##  3 Healthy             3   706     823         0.858
##  4 Healthy             4   724     823         0.880
##  5 Healthy             5   756     823         0.919
##  6 Healthy             6   764     823         0.928
##  7 Tumoral             1     1      41         0.0244
##  8 Tumoral             2    13      41         0.317
```

```
##   9 Tumoral             3    21       41          0.512
##  10 Tumoral             4    23       41          0.561
##  11 Tumoral             5    26       41          0.634
##  12 Tumoral             6    26       41          0.634
```

10 Plots of cumulative incidence of first bud production

10.1 Common themes

Theme of y-axis, and its presence or absence on the plots

```
scale_y <- scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.25),
                              labels = percent, position = "right")

hide_y <- theme(axis.text.y = element_blank(), axis.ticks.y = element_blank())

show_y <- theme(axis.text.y = element_text(color = "black", size = 14),
                axis.ticks.y = element_line())
```

Theme of x-axis, and its presence or absence on the plots.

```
scale_x <- scale_x_continuous(limits = c(0, 6), breaks = seq(0, 6, by = 2))

hide_x <- theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())

show_x <- theme(axis.text.x = element_text(color = "black", size = 14),
                axis.ticks.x = element_line())
```

Color scale for tumoral state

```
tum_scale <- list(
  scale_color_manual(values = tum_colors),
  scale_fill_manual(values = tum_colors))
```

Color scales for sleep condition

```
sleep_scale <- list(
  scale_color_manual(values = sleep_condition_colors, breaks = c("Control", "Sleep
  deprivation")),
  scale_fill_manual(values = sleep_condition_colors, breaks = c("Control", "Sleep
  deprivation")))
```

Theme shared by panels A and C

```
theme_common <- theme(
  axis.title.x = element_text(color = "black", size = 14, margin = margin(t = 10)),
  axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
                                    margin = margin(l = 15)),
  plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Presence or absence of labels on the plots

```
labs_shown  <- labs(x = "Time (weeks)",
                    y = "Cumulative incidence of\nfirst bud production",
                    title = NULL)

labs_hidden <- labs(x = NULL, y = NULL, title = NULL)
```

10.2 Panel A: interaction between sleep condition and lineage

10.2.1 HO_SPT

Cumulative incidence of first bud production by sleep condition for the lineage HO_SPT.

```
A1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ sleep_condition,
               data = FB_df %>% filter(lineage == "HO_SPT") %>%
                 mutate(sleep_condition = factor(sleep_condition, levels = c("Control",
                 "Sleep deprivation")))) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + scale_y + scale_x + sleep_scale +
  theme_bw() +
  theme(axis.text.x = element_text(color = "black", size = 14),
        legend.position = "none") +
  annotate("text", x = 5.8, y = 0.05, label = "HO_SPT",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.2.2 HO_VLN

Same method as above, but for the lineage HO_VLN.

```
A2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ sleep_condition,
               data = FB_df %>% filter(lineage == "HO_VLN") %>%
                 mutate(sleep_condition = factor(sleep_condition, levels = c("Control",
                 "Sleep deprivation")))) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + scale_y + scale_x + sleep_scale +
  theme_bw() +
  theme(axis.text.x = element_text(color = "black", size = 14),
        legend.position = "none") +
  annotate("text", x = 5.8, y = 0.05, label = "HO_VLN",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.2.3 HC_MT

Same method as above, but for the lineage HC_MT.

```
A3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ sleep_condition,
                data = FB_df %>% filter(lineage == "HC_MT") %>%
                  mutate(sleep_condition = factor(sleep_condition, levels = c("Control",
                  "Sleep deprivation")))) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_shown + theme_common + show_y + scale_y + scale_x + sleep_scale +
  theme_bw() +
  theme(axis.text.x = element_text(color = "black", size = 14),
        legend.position = "none") +
  annotate("text", x = 5.8, y = 0.05, label = "HC_MT",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

Extract legend from the plot A3 to use later on the full figure.

```
shared_treatment_legend <- cowplot::get_legend(
  A3 + theme(legend.position = "right",
               legend.text = element_text(size = 13),
               legend.title = element_text(size = 14, face = "bold")))
```

Combine A1, A2, and A3 into panel “A”

```
A <- (A1 + A2 + A3 + plot_layout(nrow = 1)) +
  plot_annotation(title = "Interaction between sleep condition and lineage
  ",
                  theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                  size = 16, margin = margin(b = 10))))
```

10.3 Panel B: by tumoral state

Cumulative incidence of first bud production by tumoral state.

```
B <- cuminc(Surv(age_followup, factor(status_competing)) ~ tum_state_early, data = FB_df)
%>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_shown + scale_y + tum_scale +
  theme_bw() +
  theme(axis.text = element_text(color = "black", size = 14),
        axis.title.x = element_text(color = "black", size = 14, margin = margin(t = 10),
        hjust = 0.91),
        axis.title.y.right = element_text(color = "black", size = 14, angle = 90, margin
        = margin(l = 15)),
        plot.margin = margin(l = 5, r = 5, t = 5, b = 5),
        legend.position = "none") +
  plot_annotation(title = "Tumoral state effect                ",
                  theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                  size = 16, margin = margin(b = 10))))
```

Extract legend from this panel to use later on the full figure.

```r
shared_tumor_legend <- cowplot::get_legend(
  B + theme(legend.position = "right",
                      legend.text = element_text(size = 13),
                      legend.title = element_text(size = 14, face = "bold")))
```

10.4 Panel C: by lineage

10.4.1 HO_MT

Cumulative incidence of tumor onset for the lineage HO_MT.

```r
C1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
              data = FB_df %>% filter(lineage == "HO_MT")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + hide_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HO_MT",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.4.2 HO_SPC

Same method as above, but for the lineage HO_SPC.

```r
C2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
              data = FB_df %>% filter(lineage == "HO_SPC")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + hide_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HO_SPC",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.4.3 HO_SPT

Same method as above, but for the lineage HO_SPT.

```r
C3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
              data = FB_df %>% filter(lineage == "HO_SPT")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + show_y + hide_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HO_SPT",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.4.4 HO_VLN

Same method as above, but for the lineage HO_VLN.

```
C4 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
               data = FB_df %>% filter(lineage == "HO_VLN")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + show_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HO_VLN",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.4.5 HC_MT

Same method as above, but for the lineage HC_MT.

```
C5 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
               data = FB_df %>% filter(lineage == "HC_MT")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_hidden + theme_common + hide_y + show_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HC_MT",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

10.4.6 HV_GAL

Same method as above, but for the lineage HV_GAL.

```
C6 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
               data = FB_df %>% filter(lineage == "HV_GAL")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_shown + theme_common + show_y + show_x + scale_y + scale_x +
  theme_bw() +
  annotate("text", x = 5.8, y = 0.05, label = "HV_GAL",
           color = "black", fontface = "bold", size = 5, hjust = 1, vjust = 0)
```

Create panel C combining plots C1 to C6.

```
C <- (C1 + C2 + C3) / (C4 + C5 + C6) +
  plot_layout(heights = c(1, 1)) +
  plot_annotation(title = "Lineage effect                ",
                  theme = theme(plot.title = element_text(
                    hjust = 0.5, face = "bold", size = 16, margin = margin(b = 10))))
```

10.5 Assembly

Add a letter tag to the first sub-plot of each panel.

```
A[[1]] <- A[[1]] + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face =
"bold"))
B[[1]] <- B[[1]] + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face =
"bold"))
C[[1]][[1]] <- C[[1]][[1]] + labs(tag = "C") + theme(plot.tag = element_text(size = 20,
face = "bold"))
```

Convert each panel into a graphical object.

```
A <- patchwork::patchworkGrob(A)
B <- patchwork::patchworkGrob(B)
C <- patchwork::patchworkGrob(C)
```

Align panels A, B, and C vertically.

```
left_column <- cowplot::plot_grid(A, B, C, ncol = 1, align = "v", axis = "r",
                                  rel_heights = c(1, 1, 1.6))
```

Shift shared legend to the right so they align with the plots they refer to.

```
shared_treatment_legend_shifted <- cowplot::plot_grid(NULL, shared_treatment_legend, ncol
= 2, rel_widths = c(3, 0.85))
shared_tumor_legend_shifted <- cowplot::plot_grid(NULL, shared_tumor_legend, ncol = 2,
rel_widths = c(0.15, 0.85))
```

Place the legends on the right of the figure.

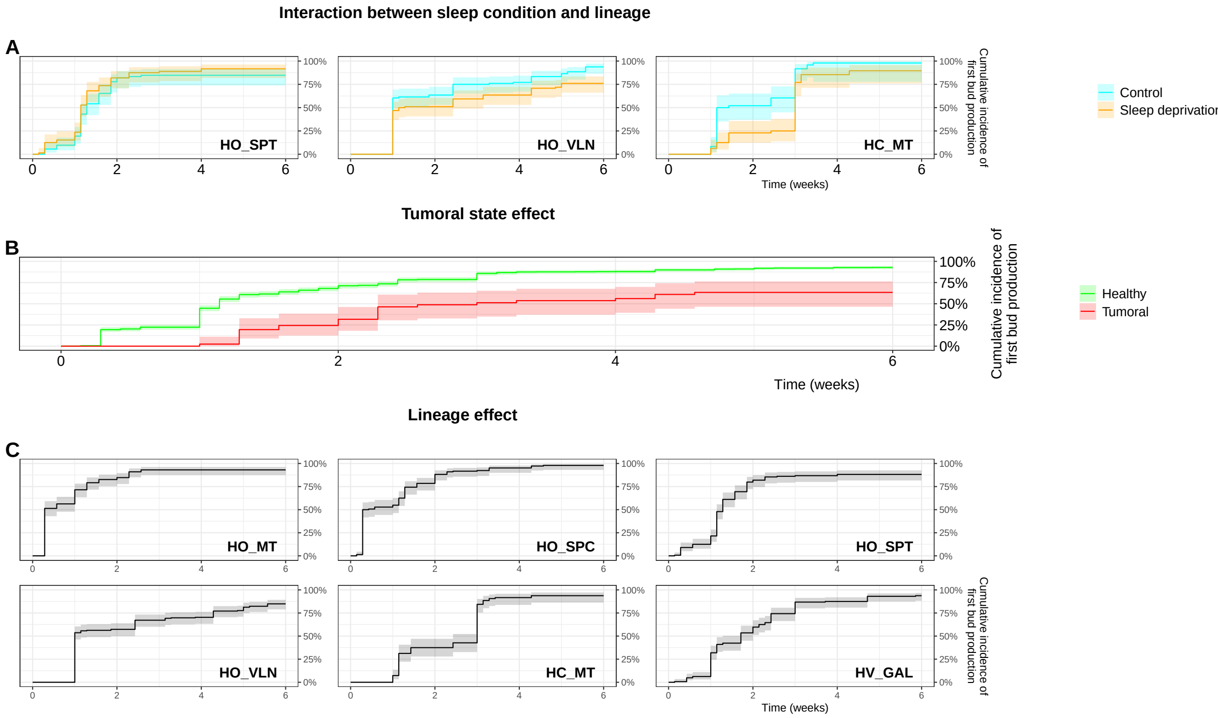
```
right_column <- cowplot::plot_grid(shared_treatment_legend_shifted,
                                   shared_tumor_legend_shifted, NULL,
                                   ncol = 1, rel_heights = c(1, 1, 1.6))
```

Combine the panels and the shared legend.

```
final_FB_panel <- cowplot::plot_grid(left_column, right_column, ncol = 2, rel_widths =
c(1, 0.14)) +
  theme(plot.margin = margin(l = 5, r = 60, t = 5, b = 5))
```

Show the final figure.

```
print(final_FB_panel)
```


Interaction between sleep condition and lineage
A
HO_SPT
HO_VLN
HC_MT
Cumulative incidence of first bud production
Time (weeks)
Control
Sleep deprivation
Tumoral state effect
B
Cumulative incidence of first bud production
Time (weeks)
Healthy
Tumoral
Lineage effect
C
HO_MT
HO_SPC
HO_SPT
HO_VLN
HC_MT
HV_GAL
Cumulative incidence of first bud production
Time (weeks)